HW7: Bayesian and Nonparametric Regression with Gaussian Processes

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1 Objective and Setup

The goal of this assignment is to compare how different regression methods learn from the Airfoil Self-Noise dataset and how their modeling assumptions change the fitted function, stability, and interpretability. The methods studied are linear regression, polynomial regression, ridge regression, kernel ridge regression with an RBF kernel, and Gaussian Process regression.

The dataset contains five input variables: frequency, angle of attack, chord length, velocity, and displacement thickness. The response variable is sound pressure level. Since the input space is five-dimensional, direct visualization is not possible. I selected `frequency` as the primary variable for plotting, fixed the other four inputs at their training-set mean values, created a grid over the observed range of `frequency`, and plotted model predictions along that slice.

All models were fit using standardized predictors. The target variable was also standardized during fitting, and predictions were transformed back to the original pressure scale before plotting and before computing mean squared error (MSE). A 70/30 train-validation split with `random_state = 42` was used throughout, and Task 4 used 5-fold cross-validation for model selection.

2 Task 1: Linear Regression and Global Modeling

Linear regression provides the baseline global model. Its fitted slice is shown in Figure 1.

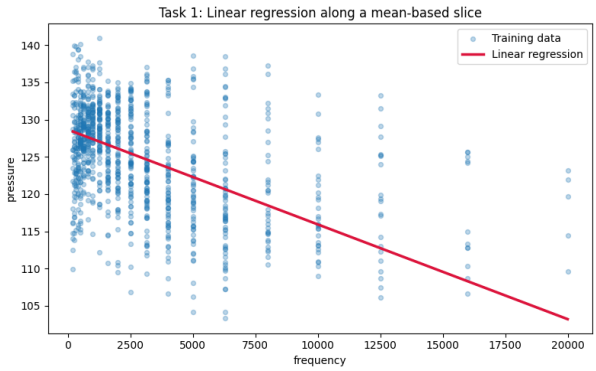

Figure 1: Linear regression predictions as frequency varies and the other features are fixed at their mean values.

The linear model obtained a training MSE of 22.827 and a validation MSE of 23.688, with validation $R^2 = 0.498$. The fitted curve is nearly a straight line, which means the model captures only a single global trend. The scatter around this line remains substantial, so the model clearly underfits the dataset. This is the main limitation of global linear regression here: it cannot represent nonlinear structure or more complex feature interactions.

3 Task 2: Polynomial Regression and Model Complexity

To increase flexibility, polynomial regression models of degrees 2 through 6 were fit and compared. Figure 2 shows the fitted slices, while Figure 3 summarizes how training and validation MSE change with degree.

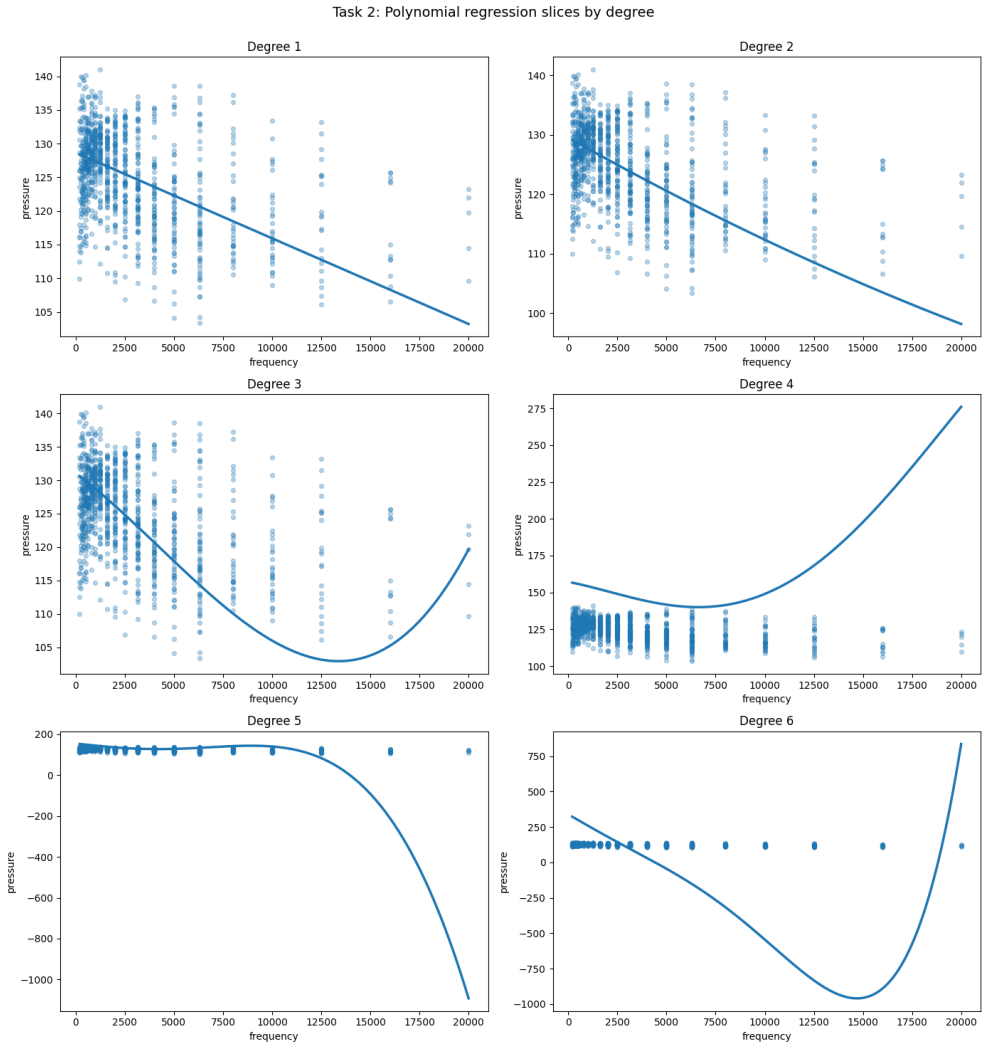

Figure 2: Polynomial regression fits for degrees 2 through 6. Included linear regression (degree 1) also here for easier comparison

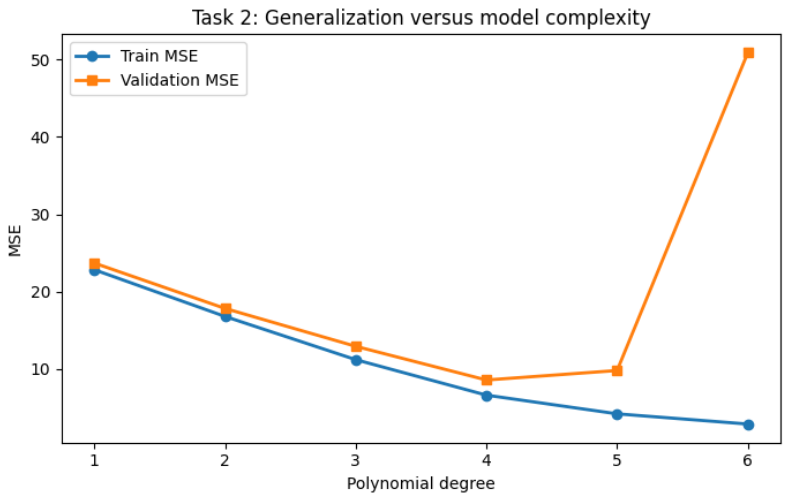

Figure 3: Training and validation MSE as model degree increases.

The validation MSE improved steadily from 23.688 at degree 1 to 17.821, 12.921, and 8.565 at degrees 2, 3, and 4 respectively. This indicates that moderate nonlinearity is necessary to explain the data. However, beyond degree 4 the behavior became unstable. Degree 5 already began to lose validation performance, and degree 6 deteriorated sharply to a validation MSE of 50.882. This is a clear example of overfitting: the training MSE kept falling, but the model generalized very poorly.

4 Task 3: Ridge Regression and Regularization

Ridge regression was applied to the degree-6 polynomial basis to study how regularization affects an unstable high-complexity model. Figure 4 compares the fitted slices for several values of λ.

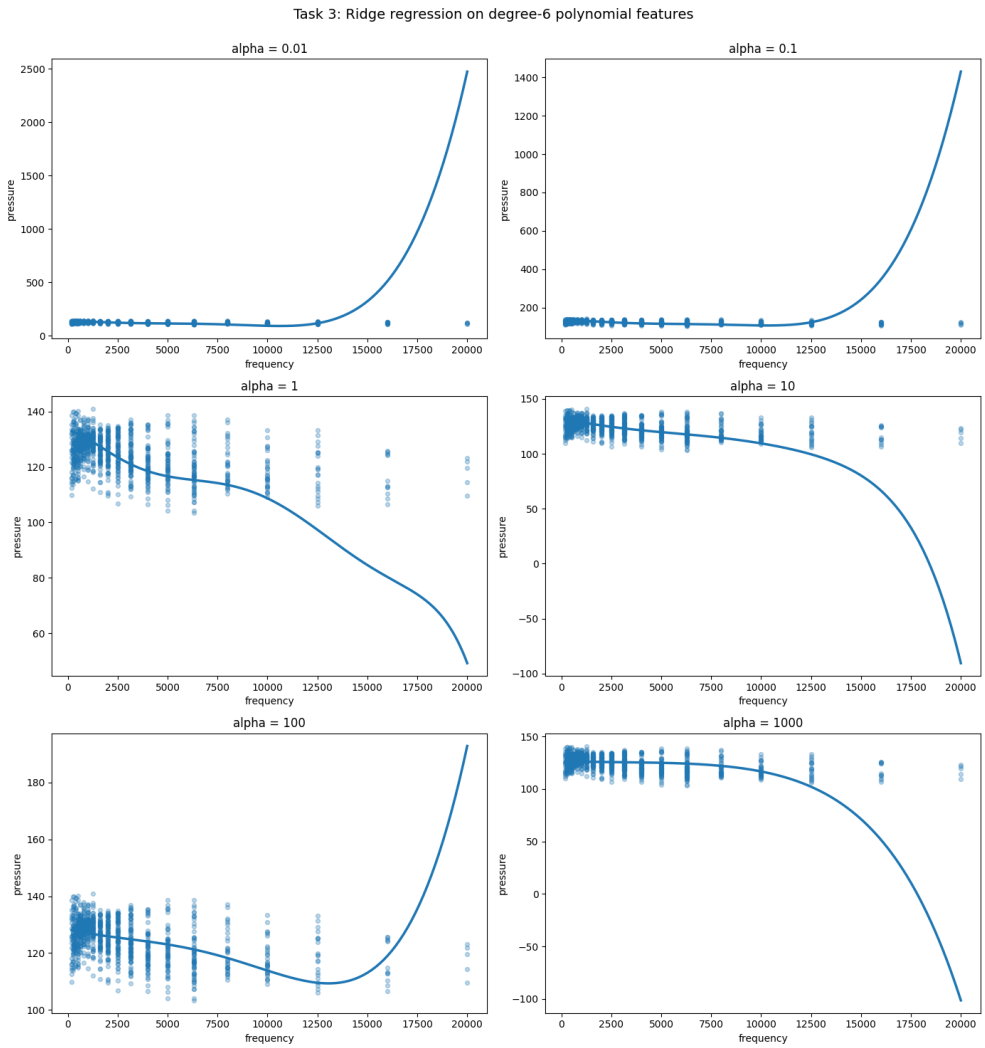

Figure 4: Degree-6 ridge regression for multiple regularization strengths.

The results show that regularization stabilizes the fitted function. With very small λ, the model still behaves like the unstable degree-6 polynomial. As λ increases, the curve becomes smoother and less extreme. In this sweep, validation MSE improved from 15.012 at $\lambda = 0.01$ to 12.169 at $\lambda = 100$, before worsening again to 15.774 at $\lambda = 1000$. This is exactly the bias-variance tradeoff: moderate regularization reduces variance and improves generalization, but excessive regularization introduces too much bias.

5 Task 4: Model Selection and Generalization

Model selection was performed using 5-fold cross-validation over polynomial degrees 1 through 6 and ridge penalties $\lambda \in \{0.01, 0.1, 1, 10, 100, 1000\}$. To avoid leakage, scaling was refit inside each fold. The mean cross-validation MSE is visualized in Figure 5, and the best ridge penalty at each degree is summarized in Figure 6.

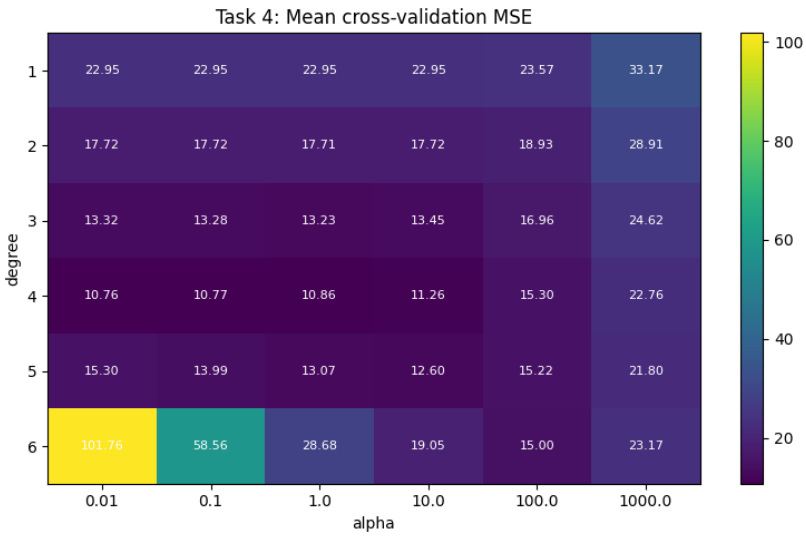

Figure 5: Mean cross-validation MSE across polynomial degree and ridge penalty.

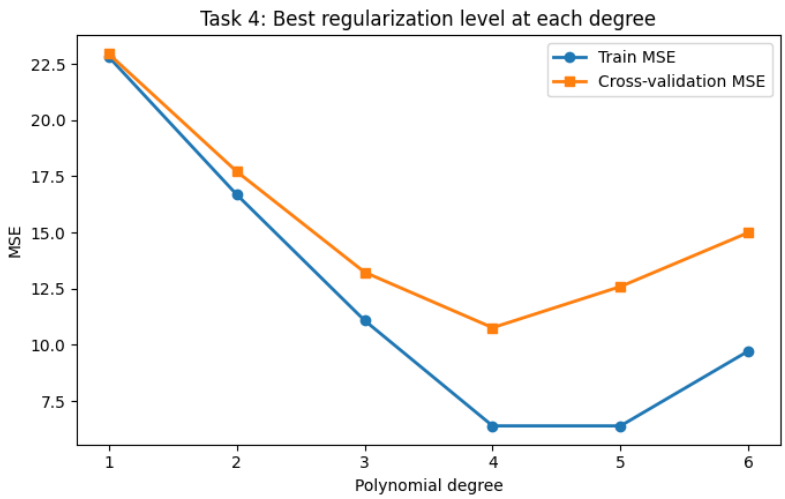

Figure 6: Training and cross-validation MSE for the best ridge penalty at each degree.

The best cross-validated model was a degree-4 ridge regression with $\lambda = 0.01$. Its mean cross-validation MSE was 10.763, and when refit on the full training split it achieved a holdout validation MSE of 8.449 with validation $R^2 = 0.821$. The training error keeps falling as degree increases, but the cross-validation error rises again for degrees 5 and 6 (an example overly complex models performing poorly). Model selection should focus on generalization rather than training fit alone.

6 Task 5: Kernel-Based Regression and Local Learning

Kernel ridge regression with an RBF kernel was then used to shift from global basis expansions to local similarity-based learning. The regularization was fixed at $\alpha = 0.01$, and the RBF parameter γ was varied. The resulting prediction slices are shown in Figure 7.

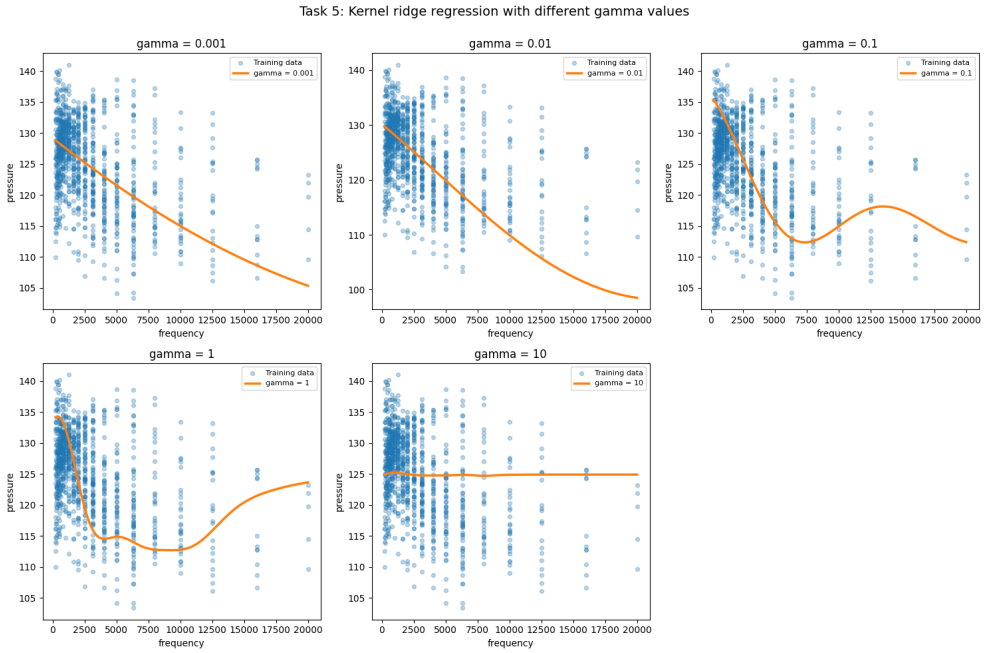

Figure 7: Kernel ridge regression for different RBF locality parameters.

The behavior changes systematically with γ. Small values produce broad smoothing that behaves almost like a global model, while large values produce highly local fits. The best setting in this sweep was $\alpha = 0.01$ and $\gamma = 1.0$, which achieved validation MSE 4.860 and validation $R^2 = 0.897$. This is substantially better than the best polynomial and ridge models, suggesting that local similarity is a better inductive bias for this dataset than a fixed global polynomial basis.

7 Task 6: Gaussian Process Regression and Uncertainty

Gaussian Process regression with an RBF kernel extends the locality idea by producing both a mean prediction and a predictive uncertainty estimate. I first examined fixed RBF length scales and then fit an optimized GP model. Figure 8 shows how validation MSE changes with a fixed length scale, and Figure 9 shows the predictive mean together with a $\pm 2\sigma$ uncertainty band.

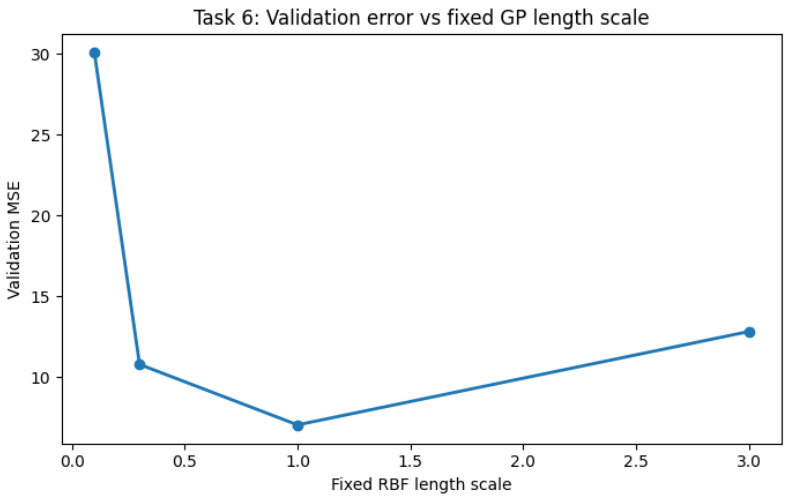

Figure 8: Validation MSE for Gaussian Processes with fixed RBF length scales.

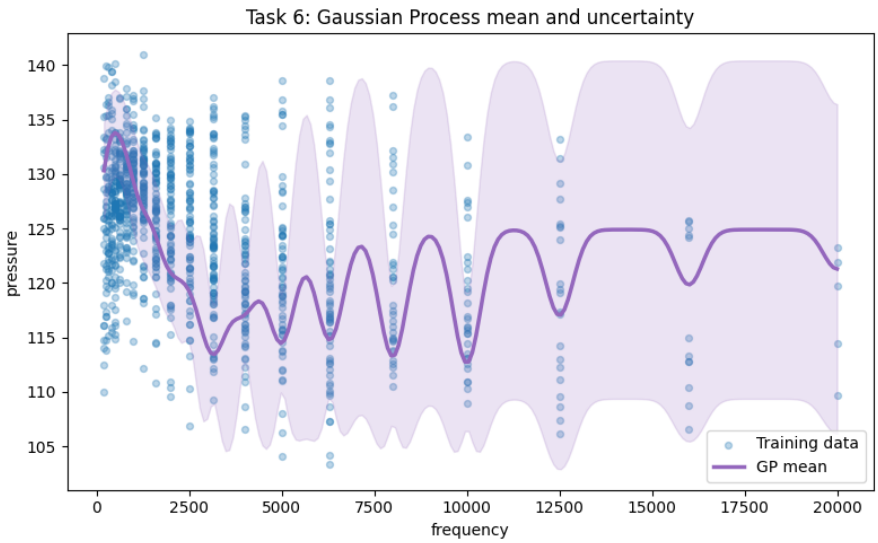

Figure 9: Optimized Gaussian Process mean prediction with $\pm 2\sigma$ uncertainty band.

Among the fixed-length-scale models, a moderate length scale near 1.0 performed best, with validation MSE 7.062. After optimization, the GP converged to the kernel

$$1.12^2 \times \mathrm{RBF}([0.113, 1.12, 0.732, 3.29, 0.482]) + \mathrm{WhiteKernel}(0.0158).$$

This optimized model achieved the strongest holdout performance of the notebook with validation MSE 4.456 and validation $R^2 = 0.906$. The uncertainty band is narrower where the slice passes through regions with stronger local support from the data and wider where that support is weaker. That is the main conceptual gain over kernel ridge regression: the GP provides uncertainty in addition to a smooth predictive mean.

8 Task 7: Comparative Analysis and Final Choice

Figure 10 overlays representative fits from each model family. Table 1 summarizes the final validation metrics.

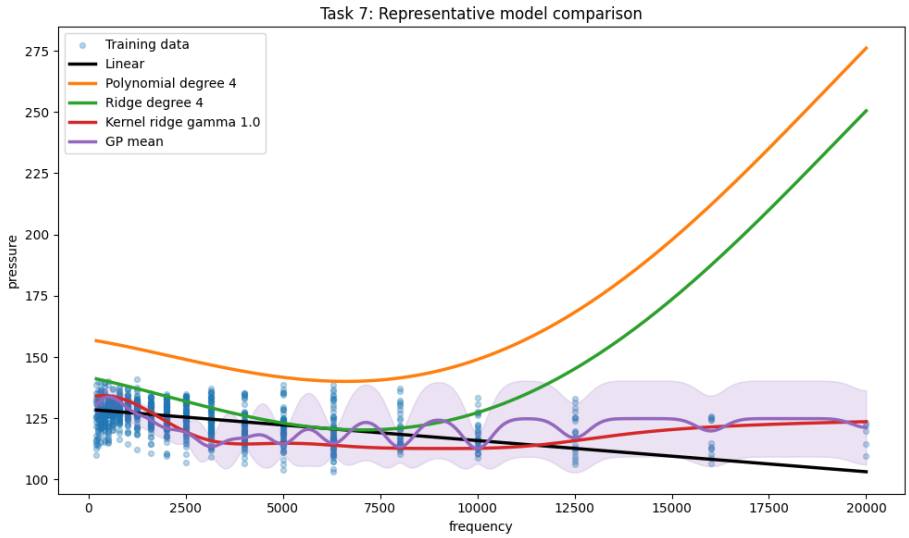

Figure 10: Representative comparison of linear, polynomial, ridge, kernel ridge, and Gaussian Process fits.

Model	Validation MSE	Validation R^2
Linear regression	23.688	0.498
Polynomial regression, degree 4	8.565	0.819
Ridge regression, degree 4, $\lambda = 0.01$	8.449	0.821
Kernel ridge, $\alpha = 0.01$, $\gamma = 1.0$	4.860	0.897
Gaussian Process	4.456	0.906

Table 1: Holdout comparison across representative model families.

The comparison reveals a clear ordering. Linear regression is too rigid and underfits badly. Polynomial regression captures useful nonlinearity, but very high degree models become unstable. Ridge regression improves stability, yet it still belongs to the same global basis expansion family. Kernel ridge improves substantially because it uses locality rather than a single global polynomial surface. The Gaussian Process performs best overall because it combines local flexibility with uncertainty quantification.

If I had to choose one model for this dataset, I would choose the optimized Gaussian Process. It achieves the lowest validation MSE and highest validation R^2, and it provides uncertainty estimates that are valuable when interpreting predictions. The tradeoff is that GP models are less transparent and more computationally expensive than linear or polynomial models. Still, on this dataset the richer modeling assumptions are justified by the clear improvement in predictive quality.

9 Conclusion

This assignment shows how regression performance depends strongly on the underlying modeling assumptions. Global linear models are too simple, unregularized high-degree polynomials are too unstable, and regularization can improve but not fully solve the limitations of global basis expansions. Local similarity-based approaches are a better fit for this dataset, and Gaussian Processes add the further advantage of predictive uncertainty. The final outcome is therefore both empirical and conceptual: the Gaussian Process is the strongest performer, and it also offers the most informative representation of what the model knows and where it is uncertain.